Semester: 3

Course Title: Data Structures

Course Code: 23CS3PCDST

Introduction: Data and Information, Data Structure, Classification of Data Structures, Data structure versus File Organization, Operations on Data Structure.

1.1 Introduction

Among the vast applications of the computer, the major application is to manage and process large amounts of data. The knowledge about representing data is the fundamental to study of computer science. The primary objective behind computer programming should be to do the calculations efficiently and to store data and retrieve data in an optimistic way. Some structures need to be devised to store, organize, and search through data. Thus, the study of structuring the data (hence data structures) and algorithms that manipulate data is the heart of computer science.

The problems solved on digital computers have become progressively large and more complex. Such a situation involves the use of appropriate data organization and efficient storage representation. So, the basic understanding of the term data structure and its importance in information systems is necessary.

1.2 Data and Information

Before starting the discussion on the data structure and its types, it would be better to understand what data is and how it is important for any decision-making process. The term data has been derived from the word datum (means that is given). Data is plural of datum. The term data refer to the value or simply set of values that are raw and unorganized. Data may be simple, random, and useless until it is organized (so called processed). Data is valuable raw material which can be in different forms such as numbers, words, alphabets, etc.

When data is processed i.e. organized, structured, and presented in a meaningful context so that it becomes useful for decision making and understanding, it becomes Information. In layman language, data and information are interchangeable terms, but technically, no conclusion can be drawn from data. Information play vital role in decision making which is obtained from data. So, indirectly data is an important entity for decision making.

As an example, Student’s marks in a subject are treated as a data item. Whereas class’s average marks or school's average marks in the same subject is information which can be calculated from the data (student's individual marks).

As another example, sales figures of different products in different cities over a time period are data items. When these sales figures are processed to find out average sale of a product in a particular city becomes information. More precisely, the graphs/plots created from sales data to present the average sales are the information.

1.3 Data Structure

We can organize data into the computer's memory in various logical ways. A data structure is a way of storing the data in computer's memory so that it can be used efficiently. Formally, a data structure is a logical/mathematical model of organization of data. The choice of data structure begins with the choice of an Abstract Data Type (ADT) such as array.

A well-designed data structure allows a variety of operations to be performed very efficiently (taking less time, less space, and a little resource).

1.3.1 Classification of Data Structures

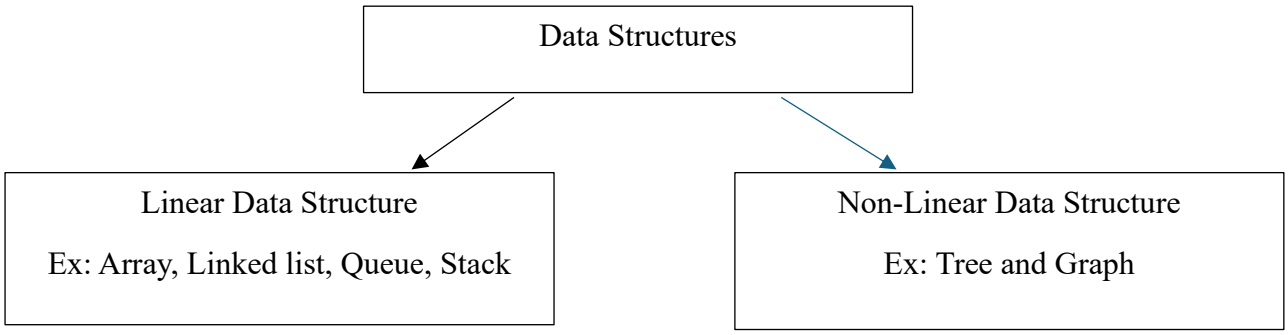
Data structures can be classified in several ways. These classifications are:

- *Linear and Non-Linear Data Structures*
- *Static and Dynamic Data Structures*
- *Homogeneous and Non-Homogeneous Data Structures*

Linear and Non-Linear Data Structures

Linear Data Structure: The elements in a linear data structure form a linear sequence. Example of the linear data structures are: Array, Linked list, Queue, Stack etc.

Non-Linear Data Structure: The elements in a non-linear data structure do not form any linear sequence. For example, Tree and Graph



Static and Dynamic Data Structures

Static Data Structure: Static data structures are those whose memory occupation is fixed. The memory taken by these data structures cannot be increased or decreased at run time. Example of the static data structure is an Array. The size of an array is declared at the compile time and this size cannot be changed during the run time.

Dynamic Data Structure: Dynamic data structures are those whose memory occupation is not fixed. The memory taken by these data structures can be increased or decreased at run time. Example of the dynamic data structure is Linked List. The size of linked list can be changed during the run time.

Other data structures like stack, queue, tree, and graph can be static or dynamic depending on, whether these are implemented using an array or a linked list.

Homogeneous and Non-Homogeneous Data Structures

Homogeneous Data Structure: Homogeneous data structures are those in which data of same type can be stored. Example of the homogeneous data structure is an Array.

Non-Homogeneous Data Structure: Non-Homogeneous data structures are those in which data of different types can be stored. Example of the non-homogeneous data structure is linked list.

Other data structures like stack, queue, tree, and graph can be homogeneous or non-homogeneous depending on, whether these are implemented using an array or a linked list respectively.

Note: There are two ways of storing linear data structure into the computer's memory. In the first way, the linear relationship between the elements of the structure is represented by sequential memory locations. For example, arrays, as elements of an array are stored in continuous/adjacent memory locations. In second way, the linear relationship between the elements of the structure is given by using pointers (addresses) e.g. linked list.

Data Structure Vs File Organization

The file organization is the study of storing the data records into the files. There are various file organization techniques like sequential file organization, random file organization, and indexed sequential file organization. These file organization techniques differ in record sequencing and retrieval methods used.

Conceptually, a data structure and a file organization are same but differ in the following aspects:

- In Implementation
- In Access Methods
- Data structures are thought of as in main memory (RAM) and file organization as in auxiliary storage (disks)

Operations On Data Structures

This section discusses the different operations that can be performed on the various data structures

Traversing: It means to access each data item exactly once so that it can be processed. For example, to print the names of all the students in a class.

Searching: It is used to find the location of one or more data items that satisfy the given constraint. Such a data item may or may not be present in the given collection of data items. For example, to find the names of all the students who secured 100 marks in mathematics.

Inserting: It is used to add new data items to the given list of data items. For example, to add the details of a new student who has recently joined the course.

Deleting: It means to remove (delete) a particular data item from the given collection of data items. For example, to delete the name of a student who has left the course.

Sorting: Data items can be arranged in some order like ascending order or descending order depending on the type of application. For example, arranging the names of students in a class in an alphabetical order, or calculating the top three winners by arranging the participants' scores in descending order and then extracting the top three.

Merging: Lists of two sorted data items can be combined to form a single list of sorted data items.

Many a time, two or more operations are applied simultaneously in a given situation. For example, if we want to delete the details of a student whose name is X, then we first have to search the list of students to find whether the record of X exists or not and if it exists then at which location, so that the details can be deleted from that particular location.